INTRODUCTION TO JENKINS PIPELINES

TYPES OF PIPELINES

1. declarative
2. scripted

Declarative pipeline

1. Declarative pipeline is a structured and easy-to-use Jenkins pipeline format.
2. It uses predefined syntax like pipeline, agent, and stages.
3. It reduces errors and is best for beginners and standard workflows.
4. It is widely used in production CI/CD pipelines.
5. Declarative pipelines are easy to use pipelines
6. In declarative pipelines we can write multiple stages as well as the multiple steps
7. Declarative pipelines start with the invocation word pipeline

```
pipeline {
   agent { label 'worker' }

   stages {
      stage('Build') {
         steps {
            echo 'Building application...'
         }
      }

      stage('Test') {
         steps {
            echo 'Testing application...'
         }
      }
   }
}
```

Scripted Pipeline

1. Scripted pipelines are hard to use pipelines
2. Generally, these pipelines start with the word node
3. And these pipelines are written in the form of script not in multiple steps so there are more chances of the errors

4. that's why we generally used these pipelines for the complex and the customized workflows.

```
node('worker') {

   stage('Build') {
      echo 'Building application...'
   }

   stage('Test') {
      echo 'Testing application...'
   }

}
```
